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About

I consider myself a quantitative urban health geographer who strives to understand the complex pathways through which urban built/natural environments (e.g., elements) influence human health and mental well-being, considering social and spatial inequalities. I advance the domain by integrating large-scale public and commercial data from various sources with state-of-the-art machine learning-based analysis and geographical information science (GIS) capabilities to provide a basis for interventions toward an inclusive, just, and human-centred urban environment. Recently, I have gained valuable insights into how street-level built and natural features, as well as micro-scale participatory interventions, could support active mobility behaviour and travel satisfaction among diverse populations. I have extensive experience in collecting data on human behaviour, attitudes, and experiences using both traditional (e.g., surveys and questionnaires) and newly emerging methods, such as public participation GIS and user-generated geo-located social media content. Additionally, I work as a self-employed GIS analyst and data scientist, having taught and mentored over a hundred individuals from various academic fields. I am also actively contributing to the body of global knowledge by publishing high-impact articles in international peer-reviewed journals and have reviewed several manuscripts for internationally reputable journals.

Area of Expertise

- Public participation science
- Human behaviour and behavioural change
- Perceived accessibility and walkability
- Urban mobility and environmental exposure
- Environmental psychology and health geography
- Spatial- data science & GIS
- Advanced statistical modelling
- Deep learning for time-series prediction and NLP
- Data visualization & web-based mapping/ Cartographic Visualization
- Spatial modelling and multimodal accessibility

Education

2020 – 2021

Applied Data Science

University of Tehran - One-year Professional Training Certificate

Subject Covered: Statistics & Probability; SQL; Machine Learning; Deep Learning; Text Mining, Big Data Analytics; Spatial Data Science and Geo-Intelligence, Python Programming; R Programming.

2017 – 2020

MSc Geography and Urban Planning

Department of Human Geography at Kharazmi University; No.43, South Mofatteh Ave, Tehran, Iran.

Dissertation Title: Linking multiple nature exposure types in daily urban trips to human health and mental well-being: A mixed-method approach. (Supervisor: Dr.

Tajeddin Karami and Co-supervisor: Dr. Ahmad Zanganeh)

Subject Covered: Geographic Information System (GIS); Remote Sensing; Google Street view imagery; Participatory GIS; Spatial Data Science; Urban Nature Exposure Assessment.

2012 – 2016

Bachelor of Geography and Urban Planning

Department of Geography at University of Maragheh; Daneshgah Boulevard, Madar Square, Daneshgah Blvd, Maragheh, Iran.

Subject Covered: Urban Land Use Planning; Planning theory, Urban Geography; RS and GIS; Urban and Regional Planning; Urban Sustainability; Participatory Planning; Project Management.

Research experience

June, 2025 –
Continuing (Part-time
project)

GIS Developer

Organisation: *Routaa* (Location-based startup) – PART of CHABOK Holding – www.routaa.com

Responsibilities:

My main responsibility in this project is to collect, maintain, and develop efficient GIS databases, as well as assist in designing and developing web-based applications for both desktop and mobile access.

November, 2024 –
Continuing

Data Scientist & Big Data Analyst

Organisation: *Shahrpat*- Institute for public opinion monitoring. Tehran Urban Research and Planning Centre (TURP) – www.rpc.tehran.ir

Responsibilities:

- Data collection from online resources (e.g., geo-tagged social media contents, blogs, APIs, etc.), pre-processing and conducting statistical and AI driven models to understand people's experience and behaviour in urban environments/nature visitation
- Co-designing questionnaires, surveys and possible participatory GIS techniques for data collection
- Providing data-driven insights to policymakers and stakeholders for evidence-based urban planning and decision-making (e.g., through interactive dashboards, executive summaries, stakeholder presentations and workshops)

October, 2023 –
October, 2024

Senior Geospatial Data Scientist

Organisation: Farnahd Urbanism & Architecture Consultant firm. Part of G-15 project – The green 15-minute city: nature contact in daily life experience, funded by Tehran municipality. (*project lead: Dr. Mahmoud Rahimi and co-lead: Hossein Eghbali*)

Responsibilities:

- Spatial modelling of active travel accessibility using open-source GIS data
- Modelling spatial and temporal variation in greenery exposure during daily urban trips (GPS tracking + open/commercial high-resolution satellite data)

June, 2021 –
October, 2023

- Applying deep Convolutional Neural Networks (CNNs) to segment **geo-located** Google Street View imagery
- Quantifying and harmonising composite greenspace metrics for greenery exposure assessment
- Developing an interactive online tool for communicating and visualising project outputs
- Maintaining a GitHub repository for project scripts and documentation
- Co-authoring policy brief/report

Urban Planer

Organisation: Farnahd Urbanism & Architecture Consultant firm. Part of COCHALLENGE project - *co-creation of approaches and solutions for the sustainable development in challenged neighbourhoods*, funded by Karaj municipality. (project lead: Dr. Afshin Kokabi and co-lead: Mahsa Goldar)

Responsibilities:

- Collecting data through primary survey, focus group discussions and Public Participation GIS (PPGIS)
- Creation of new data using satellite images, OSM, survey data, social media data, etc.
- Stakeholder meetings arrangement at the district level
- Statistical analysis
- Contributing to the Action Plan (AP) and feasible study

Sub-project (NO. 1) - co-designing **green alley**: pathway to sustainable city (**GREEN PATH**). Part of tactical urbanism solutions

Responsibilities:

- Organising public participation events
- Participating in brainstorming sessions
- Engaging with potential sponsorships and partners
- Co-designing and deploying a small-scale Public Participation GIS (PPGIS) survey to collect place-based insights

Sub-project (NO. 2) - co-designing **colourful alley**: pathway to child-friendly city (CFC). Part of tactical urbanism solutions

Responsibilities:

- Organising public participation events
- Participating in brainstorming sessions
- Engaging with potential sponsorships and partners
- Co-designing and deploying a small-scale Public Participation GIS (PPGIS) survey to collect place-based insights

Sub-project (NO. 3) - co-designing **active park**: toward gender equality (APA). Part of tactical urbanism solutions

Responsibilities:

- Data collection through spatial tracking of males vs. females' behaviour in Afraz Park
- Collecting data on female visitors' preferences through a survey/questionnaire
- Organising public participation events

September, 2019 –
December, 2021

Research Assistant

Organisation: Department of Human Geography (DHG), Kharazmi University (KhU). Working on “Urban ecological resilience assessment (UERA)”. Funded by the greater Tehran municipality (GTM). (PI: Dr. Tadjeddin Karami and Co-PI: Dr. Farzaneh Sasanpour)

Responsibilities:

- Reading and summarising policy documentation and previous ecological resilience-related projects
- Data collection through survey, digitising and real-time data sources
- Segmentation and classification of remote sensing images
- Qualitative validation of GIS models using questionnaires and interviewing techniques

Publications and Invited talks

Published Journal Articles

[4] **Badakhshan**, B., Sharifi, A., & Ramezani, S. (2025). The 15-minute city in the Global South: Modeling spatial accessibility and measuring social equity across seven major Iranian cities. *Journal of Urban Mobility*, 8, 100168. <https://doi.org/10.1016/j.urbmob.2025.100168>

Impact factor: **6.1** - Citation count: **NA**

[3] **Badakhshan**, B., Sharifi, A., Karami, T., & Bavarsad, F. S. (2025). Urban heat in a very hot region: Exploring spatial inequality in exposure to extreme heat hazard in Tehran, Iran. *Urban Climate*, 61, 102480. <https://doi.org/10.1016/j.uclim.2025.102480>

Impact factor: **6.9** - Citation count: **3**

[2] **Badakhshan**, B., Sharifi, A., & Karami, T. (2025). Is life green on the other half? Linking urban green infrastructure to socio-economic inequality and spatial segregation in Tehran, Iran. *Applied Geography*, 177, 103562. <https://doi.org/10.1016/j.apgeog.2025.103562>

Impact factor: **5.4** - Citation count: **10**

[1] Arvin, M., Jalaei, M., Taheri, J., **Badakhshan**, B., Ghane, M., & Sharifi, A. (2025). Examining the relationship between urban form and social inequality: A neighborhood-level analysis. *Applied Geography*, 176, 103532. <https://doi.org/10.1016/j.apgeog.2025.103532>

Impact factor: **5.4** - Citation count: **12**

Under review papers

[1] **Badakhshan**, B., and Rinne, T. The impact of street-level greenery exposure on people's everyday mobility pattern and travel behavior in urban environments. Target Journal: *Transport Geography*

Working papers (to be submitted)

[2] Javadpoor, M., **Badakhshan**, B., & Roosta, M. Global assessment of inclusive X-minute cities. Target Journal: *Nature Communications*.

[1] **Badakhshan**, B., Scale effects in understanding built environment influence on active school travel. Target Journal: *Transport Geography*

Invited talks

[1] **Badakhshan, B.** (April 13, 2021). Modelling urban green infrastructure inequality using spatial methods and technologies. Tehran Research and Planning Center (TRPC), Tehran municipality.

[2] **Badakhshan, B.** (March 18, 2021). Once informal settlement and now formal urban area: Do you want to know *Rezashahr* better? interview with *hezaaroyekshahr* magazine. [available at <http://bitly.ws/DHqp>]

Honours and Awards

July, 2016

1. Selected to participate in the 21st University Students' Scientific Olympiad in 2016, *Geographical Science*. National Organisation of Educational Testing (NOET)
2. Best student of MSc in **Geography & Urban Planning**, Department of Human Geography, Kharazmi University

Relevant Professional Training

Jun 2023 – Continuing

Name: **Data Scientist Career Track**

Organization: 365 Data Science

February, 2023 – April, 2023

Name: **Advance webGIS programming**

Organization: GISPlus

February, 2023

Name: **Social Media Data Analysis**

Organization: K. N. Toosi University of Technology (KNTU)

October, 2022 – November, 2022

Name: **Spatial Data Science: The New Frontier in Analytics**

Organization: Esri

July, 2022

Name: **Python A-Z™: Python for Data Science with Real Exercises!**

Organization: Udemy

May, 2021

Name: **R Programming: For Data Science with Real Exercises**

Organization: Udemy

May, 2019

Name: **Google Earth Engine (GEE)**

Organization: Sharif University of Technology (SUT)

Teaching Experience

September, 2022 – Continuing

Founder; Geoanalytica – self-employment

- GIS Consultancy
- Design and deliver online training courses on GIS & RS software, and geographic data science programming skills (GEE, R, Python)
- Online coaching sessions for beginner and more advanced GIS professionals
- Free videos and step-by-step tutorials for GIS enthusiasts

October, 2020 – Continuing

Instructor; Remote Sensing and Geographic Information System Academy (girs.ir).

- Designing and delivering online courses and masterclasses on remote sensing and spatial analysis

March, 2019 –
May 2020 (Teaching
Assistant)

Demonstrator; *Department of Geography, Kharazmi University.*

- GEO2-1811009 **Spatial Thinking with GIS** (MSc, 10 students)
- GEO1-1812570 **Application of Remote Sensing and GIS in Urban Planning** (undergraduate course, 24 students)

Student Mentorship Experience

MSc (mentoring data
analysis and software
training)

1. Maysam Kamari, MSc student in Rural Geography, Department of Geography, Kharazmi University.
2. Yasir Hasanian Zawiya, MSc student in Urban Geography, Department of Geography, Kharazmi University.
3. Elham Javanbakht, MSc student in Urban Geography, Department of Geography, Kharazmi University.
4. Mohammad Kazem Adeli, MSc student in Urban Geography, Department of Geography, Kharazmi University.
5. Zahra Nasseri Akbar, MSc student in Biostatistics, Department of Biostatistics, Shahid Beheshti University (SBU).

PhD Support

1. Majid Akbari, PhD student in Urban Geography, Department of Geography, Payame Noor University.
2. Fatemeh Salehipour Bavarsad, PhD student in Architecture, Department of Civil, Building Engineering and Architecture, Università Politecnica delle Marche.
3. Sepideh Shafiei, PhD student in Sustainability Science, Institute of Geography and Spatial Planning, University of Lisbon.

External Engagements

- | | |
|---------------|---|
| October, 2016 | Traffic Surveyor; Organization: Tehran Transportation and Traffic Organization |
| October, 2011 | Volunteering for Census Taking; Organisation: Statistical Centre of Iran (SCI) |

Reviewer

I frequently review manuscripts for **multi-disciplinary and specialised** journals published by Elsevier, Springer and MDPI, including [Scientific Reports](#), [Sustainable Cities and Society](#), [Theoretical and Applied Climatology](#), Urban Forestry & Urban Greening, [Geoforum](#), and Journal of Urban Mobility.

Language Skills

Persian	Native or bilingual proficiency
Azerbaijani	Native or bilingual proficiency
English	Listening: C1 Reading: C1 Writing: C1 Spoken: C1
Turkish	Listening: C1 Reading: B2 Writing: B2 Spoken: C2

Technical Skills & Tools

Programming (selected packages)	▪ R/R Studio (e.g., sf, rgdal, rgeos, tmap, spatialreg, terra, r5r, ggplot2) ▪ Python (e.g., rasterio, Fiona, pandas, numpy, scikit-learn, matplotlib, scipy, geopandas, network, osmnx, pandana, pydeck, folium, H3) ▪ MATLAB ▪ JavaScript (e.g., leaflet and Google Earth Engine API)
GIS Software	▪ ArcGIS 10.x/pro/ArcGIS Online ▪ QGIS
Remote sensing	▪ Google Earth Engine ▪ ENVI ▪ SNAP ▪ ERDAS ▪ eCognition
Statistical Packages	▪ SPSS ▪ Minitab ▪ Stata
Data visualisation & Interactive Mapping	▪ R (Shiny) ▪ Streamlit ▪ Power BI & DAX ▪ Tableau ▪ ArcGIS StoryMaps ▪ Leaflet ▪ Pydeck ▪ Kepler.gl ▪ Mapbox
Databases	▪ PostgreSQL ▪ MySQL ▪ ArcGIS server
Qualitative data analysis software	▪ Nvivo ▪ Mozdeh
Deep learning Environment (for Computer Vision & Sequential Modeling)	▪ TensorFlow ▪ PyTorch ▪ PyTorch Geometric ▪ Transformers ▪ YOLOv4-v8
Version Control	▪ GitHub/GitLab ▪ Google Colab
Graphic Design Software	▪ Adobe Photoshop ▪ Adobe InDesign ▪ Adobe Illustrator

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